جامعة الامير سلطان
PRINCE SULTAN UNIVERSITY

Topic 1
Introduction to AI and Data Science

Instructor: Dr. Heyfa Ammar

Term 261

Prince Sultan University

Do not distribute or share any slide without permission of the author. Slides Courtesy: Prof. Anis Koubaa

Topic Overview

Topic 1

Introduction to Data Science and AI

Outline

- What is AI?
- Applications of AI
- What is Data?
- What is Data Science?
- Intersection AI-Data Science
- Applications of Data Science
- Data analytics

Intro to AI

Projected Value Creation

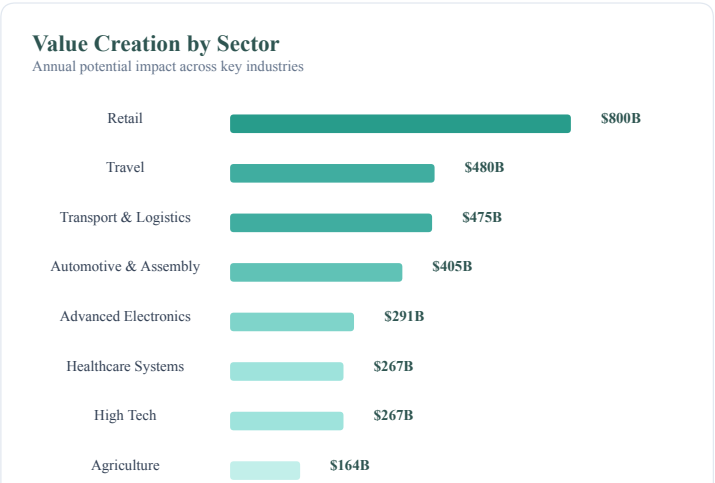
By the year 2033

$13-22 Trillion

Generative AI Impact

Includes **$3-4 trillion** from generative AI technologies annually.

[Source: McKinsey Global Institute]



Can you think of an industry that AI will not have an impact on in the next years?

What is AI?

How to get computers to think and learn like humans.
This field has been there since the early days of computing, but has sped up over past few years due to better hardware and data processing.

What is AI?

Definition by High-Level Expert Group on Artificial Intelligence (EU)

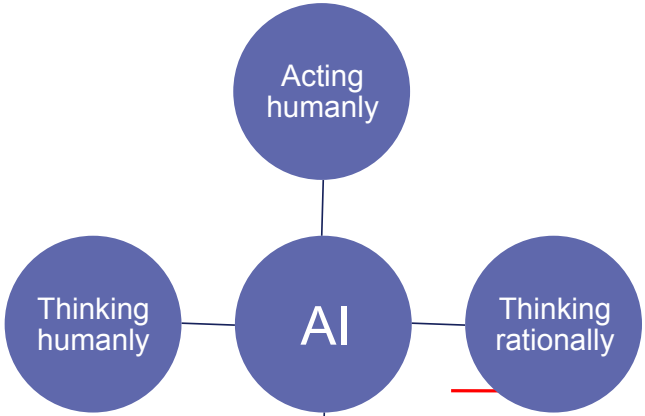
Artificial intelligence (AI) systems are **software** (programs) (and possibly also hardware) **systems** designed by humans(2) that, given a **complex goal**, act in the physical or digital dimension by **perceiving their environment through data acquisition**, **interpreting the collected structured or unstructured data**, **reasoning on the knowledge**, or processing the information, derived from this data and **deciding the best action**(s) to take to **achieve the given goal**. AI systems can either use symbolic rules or learn a numeric model, and they can also adapt their behaviour by analysing how the environment is affected by their previous actions

1) ai gets the data
2) interpret/understands the data
3) make a relationship from the data and goal (how to achieve a goal from data)

Definition by Austrian Council for Robotics and Artificial Intelligence (AT)

"... artificial intelligence (AI) refers to **computer systems** that **exhibit intelligent behavior**, i.e., that are capable of performing tasks that **in the past required human cognition and decision-making skills**. Artificial intelligence-based systems **analyze their environment** and **act autonomously** to **achieve specific goals**. ... They operate through rule knowledge created by experts or based on statistical models derived from data (machine learning, e.g., deep learning). The term AI includes both pure software, but can also include hardware, such as in the case of autonomous robots. ..."

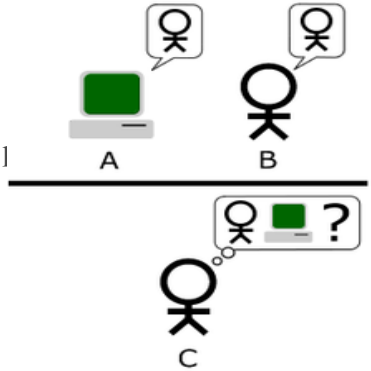
How to Achieve AI?



Acting Humanly: The Turing Test

http://en.wikipedia.org/wiki/Turing_test

- To be intelligent, a [illegible] a human

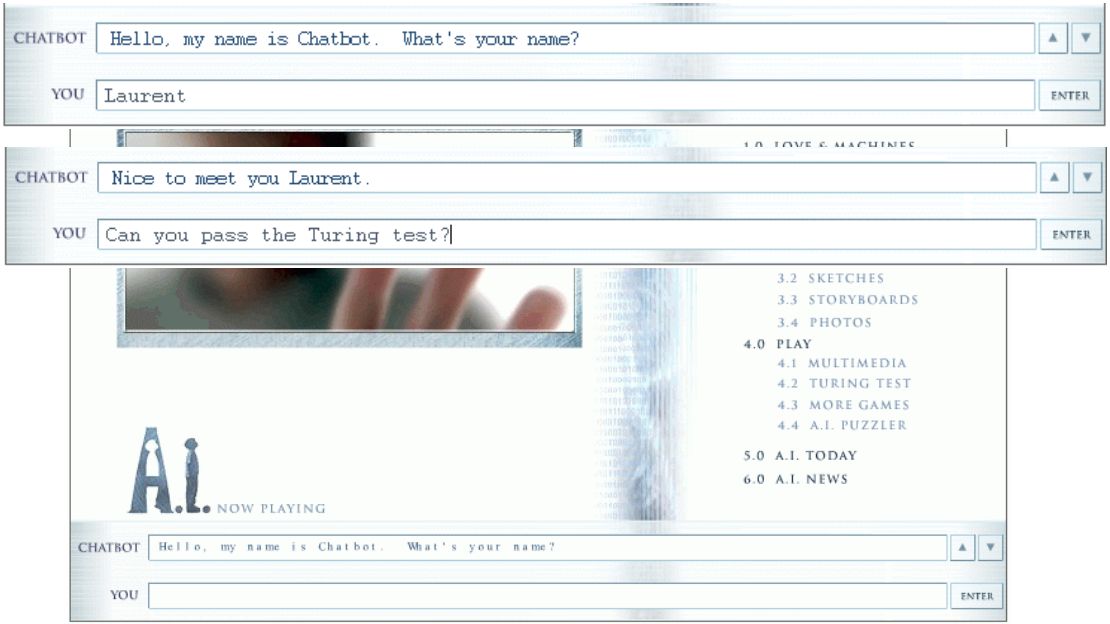


Alan Turing
1912-1954

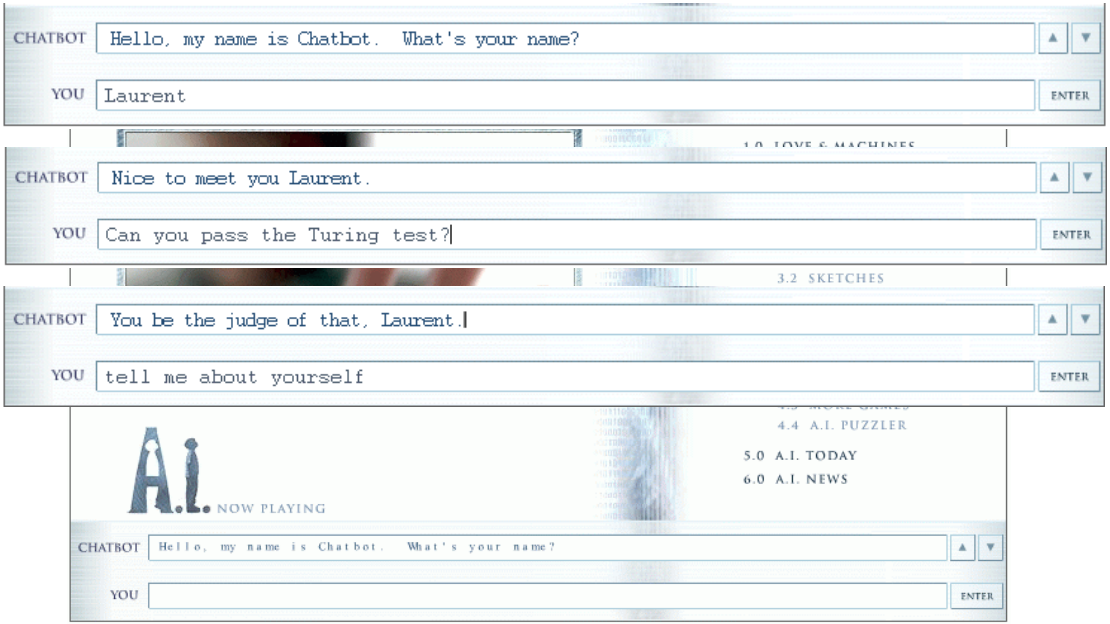
The Turing Test - Example

The Turing Test - Example

The Turing Test - Example



The Turing Test - Example



The Turing Test - Example

Can YOU pass the turing test ?

https://turingtest.live

Acting Humanly

- To pass the Turing test, the computer/robot needs:
  - **Natural language processing** to communicate successfully.
  - **Knowledge representation** to store what it knows or hears.
  - **Automated reasoning** to answer questions and draw conclusions using stored information.
  - **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

  - These are the main branches of AI.

Thinking Humanly

- Real intelligence requires thinking think like a human !
- First, we should know how a human think
- Then, we can build programs and models that think like humans

Thinking Rationally

- Instead of thinking like a human : think rationally.
- Find out how correct thinking must proceed: **the laws of thought.**
- Aristotle syllogism: “Socrates is a man; all men are mortal, therefore Socrates is mortal.”
- This initiated logic: a traditional and important branch of mathematics and computer science.

Acting Rationally

- Rational agent: acts as to achieve the best outcome
- Logical thinking is only one aspect of appropriate behavior: reactions like getting your hand out of a hot place is not the result of a careful deliberation, yet it is clearly rational.
- Sometimes there is no correct way to do, yet something must be done.
- Instead of insisting on how the program should think, we insist on how the program should act: we care only about the final result.
- Advantages:
  - more general than “thinking rationally” and more
  - Mathematically principled; proven to achieve rationality unlike human behavior or thought

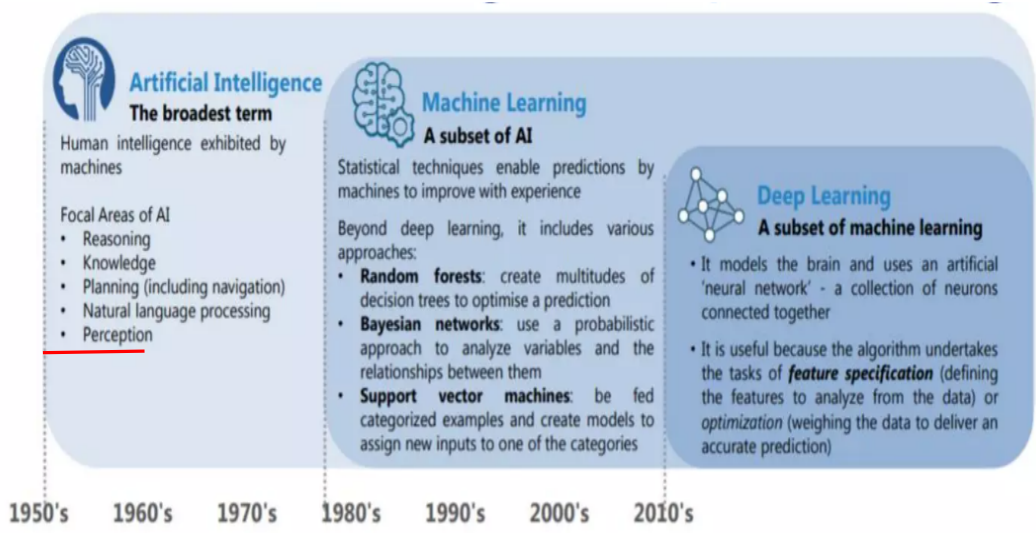
Acting Rationally

This is how birds fly

Humans tried to mimic birds for centuries

This is how we finally achieved “artificial flight”

AI, Machine learning and Deep Learning

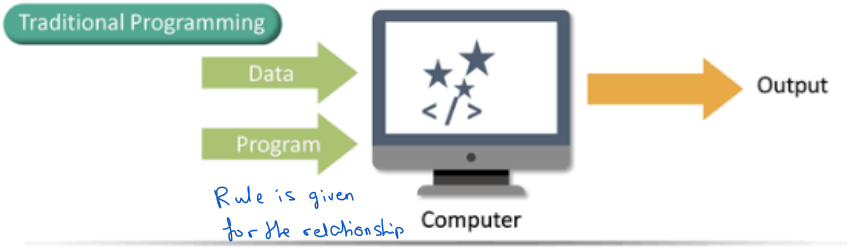


Artificial intelligence (AI) is an umbrella term for different strategies and techniques you can use to make machines more humanlike. AI includes everything from smart assistants to self-driving cars.

Machine learning (ML) is one among many other branches of AI. ML is the science of developing algorithms and statistical models that computer systems use to perform complex tasks without explicit instructions. The systems rely on patterns and inference instead. Computer systems use ML algorithms to process large quantities of historical data and identify data patterns.

While machine learning is AI, not all AI activities are machine learning.

Traditional Programming vs. Machine Learning



In both situations I dont have the output, in traditional programming I can give inputs for the program to give the output in AI, I have 1 input and some data but I cant get the output as I have multiple unknown variables so the AI approximates the output while trying to minimize the error

https://www.sketchbubble.com/en/presentation-machine-learning.html

Applications of AI

1. Healthcare and Medicine
- Disease diagnosis from medical imaging
- Drug discovery and development
- Personalized treatment recommendations
- Medical record analysis and management
- Remote patient monitoring
- Predicting patient outcomes

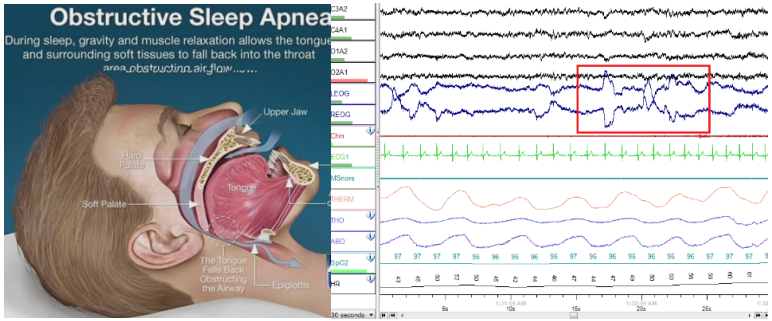
2. Business and Finance
- Fraud detection
- Algorithmic trading
- Customer service chatbots
- Risk assessment
- Market analysis and forecasting

3. Security and Surveillance
- Facial recognition
- Anomaly detection
- Threat detection
- Cybersecurity
- Network security monitoring
- Access control systems

4. Environmental Protection
- Climate modeling
- Wildlife conservation
- Energy consumption optimization
- Natural disaster prediction
- Pollution monitoring
- Resource management

...

Applications of AI



Smart agriculture

Crowd management

Applications of AI by CS316 Students, Term 241

AI Child Bracelet - Assistive Wearable for Children with ASD & ADHD

Overview

The AI Child Bracelet is a wearable device designed to assist children with Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD) during hyperfocus episodes. By using a machine learning-based voice recognition system, it detects specific voices (e.g., caregivers or teachers) and provides gentle vibration cues to help redirect the child's attention, improving situational awareness without forceful disruptions.

SuhailChatbot

SuhailChatbot is an AI-powered conversational assistant designed to share and promote knowledge about Saudi Arabia's rich culture. This chatbot provides insights into traditional food, festivals, historical landmarks, and much more, fostering cultural awareness and engagement.

What is Data?

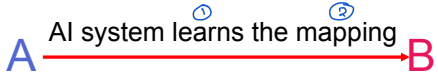
What is Data?

Example of a table of data (dataset)

data sets (Heres 2)

size of house (square feet)		price (1000$)
523		115
645		150
708		210
1034		280
2290		355
2545		440
A		B

build an AI system or machine learning system to help you set prices for houses, or figure out if a house is priced appropriately.



What is Data?

Example of a table of data (dataset)

1 feature was added

size of house (square feet)	# of bedrooms	price (1000$)
523	1	115
645	1	150
708	2	210
1034	3	280
2290	4	355
2545	4	440

A B

build an AI system or machine learning system to help you set prices for houses, or figure out if a house is priced appropriately.

A → AI system learns the mapping → B

more features gets me more precise answers when there's a correlation or when its connected to the chosen data

Sources of Data

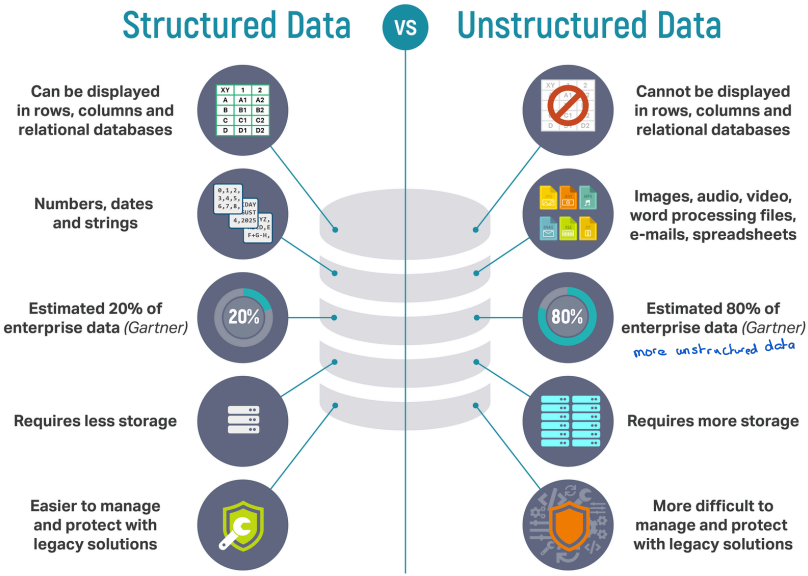
- Discussion

Types of Data

- **Structured Data**: Data that follows a predefined format, typically stored in tables (e.g., databases, spreadsheets). ex: excel
- **Unstructured Data**: Data without a formal structure (e.g., text, images, videos).
- **Semi-Structured Data**: Has some organizational properties but lacks a rigid structure (e.g., JSON, XML).
  not fixed for all data
- **Big Data**: Large-scale data that cannot be processed efficiently with traditional tools (e.g., Hadoop, Spark).

STRUCTURED vs. UNSTRUCTURED DATA

Data	Number of timepoints	Number of nodes
Antibiotics	56	386
Preterm births	216	318
Housing prices	254	944
Bikesharing	24	819
Global patterns	500	51



Audio

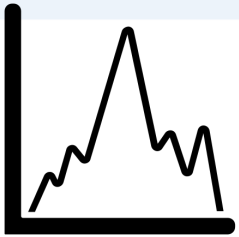
Image

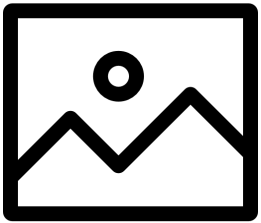
Text

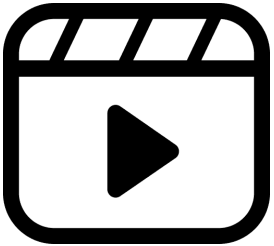
https://lawtomated.com/structured-data-vs-unstructured-data-what-are-they-and-why-care/

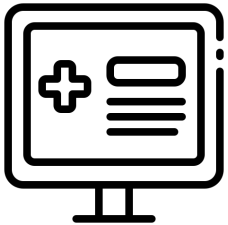
Sequence Data

Changing the order, makes it not understandable
* sequence is important (one after the other)



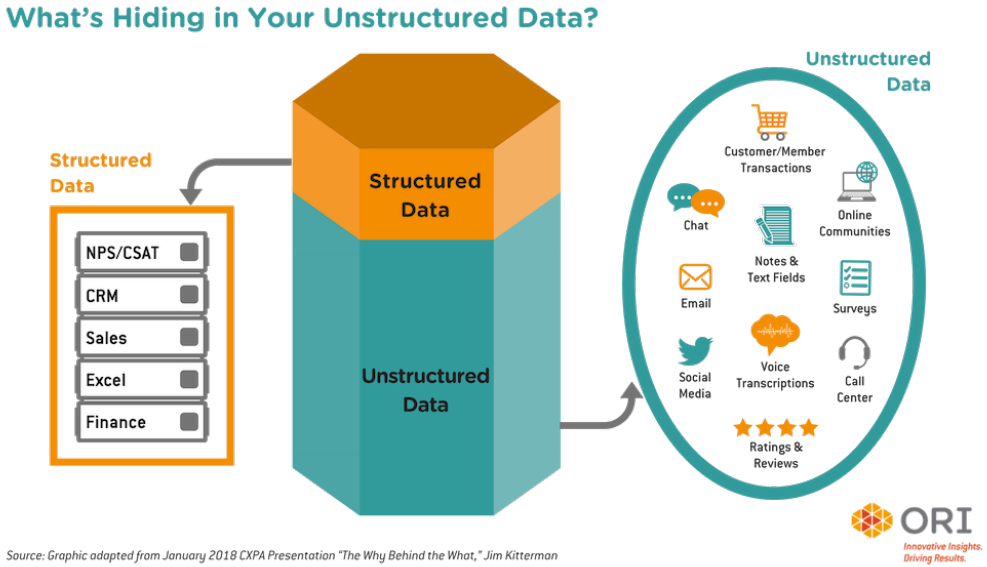


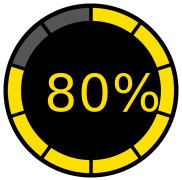




STRUCTURED vs. UNSTRUCTURED DATA

EXAMPLE





https://www.oriresults.com/articles/blog-posts/whats-hiding-in-your-unstructured-data/

Semi-Structured Data

{JSON}

json Copy code

```json
{
  "student": {
    "first_name": "Ahmad",
    "last_name": "Al-Farhan",
    "nationality": "Saudi",
    "university": "Prince Sultan University",
    "student_id": "PSU123456",
    "major": "Computer Science",
    "year": 3,
    "courses": [
      {
        "course_code": "CS101",
        "course_name": "Introduction to Computer Science",
        "grade": "A"
      },
      {
        "course_code": "MATH203",
        "course_name": "Calculus II",
        "grade": "B+"
      }
    ]
  }
}
```

xml Copy code

```xml
<student>
    <first_name>Ahmad</first_name>
    <last_name>Al-Farhan</last_name>
    <nationality>Saudi</nationality>
    <university>Prince Sultan University</university>
    <student_id>PSU123456</student_id>
    <major>Computer Science</major>
    <year>3</year>
    <courses>
        <course>
            <course_code>CS101</course_code>
            <course_name>Introduction to Computer Science</course_name>
            <grade>A</grade>
        </course>
        <course>
            <course_code>MATH203</course_code>
            <course_name>Calculus II</course_name>
            <grade>B+</grade>
        </course>
    </courses>
</student>
```

Big Data

- **Definition:**
  - Big Data refers to large-scale datasets that are too complex, large, and dynamic for traditional data processing tools to handle efficiently. so many features
- **Challenges:**
  - **Volume:** Massive amounts of data.
  - **Velocity:** High speed at which data is generated.
  - **Variety:** Different types of data (structured, unstructured, semi-structured).
  - **Veracity:** Uncertainty of data quality.

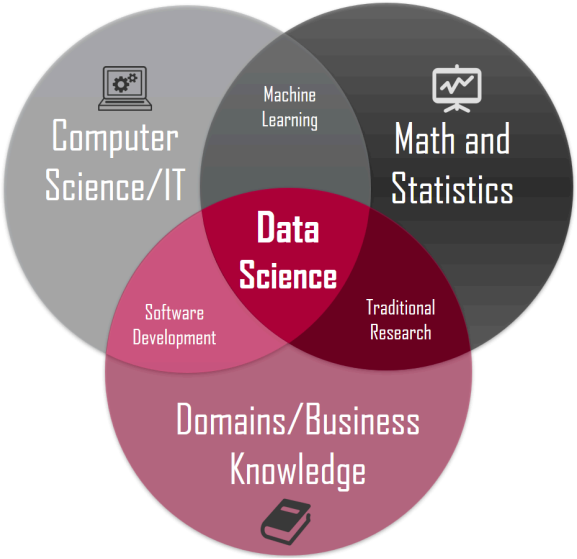
5 attributes of big data

Reference: https://www.javatpoint.com/big-data-characteristics

What is Data Science?

What is Data Science

- **Definition:** Data Science is an **interdisciplinary** field that uses scientific methods, algorithms, and systems to extract knowledge and insights from **structured** and **unstructured data**.
- **Core Components:** data science needs AI after building the model where AI acts for itself
  - Statistics,
  - Machine Learning, (AI)
  - Data Engineering,
  - Domain Expertise, and
  - Data Visualization.



https://towardsdatascience.com/introduction-to-statistics-e9d72d818745

Data Science vs. AI

What each field focuses on, and how they connect

Data Science

Goal: Extract insight from data what will this data tell me?
Output: Reports, dashboards, models
Core skills: Stats, data engineering, ML, communication
Key question: *"What does the data tell us?"*

Artificial Intelligence

Goal: Build systems that act intelligently act like a human
Output: Autonomous predictions & actions from the predictions, we take actions
Core skills: Algorithms, ML/DL, optimization, deployment
Key question: *"How can a machine act on its own?"*

Example: Reducing Hospital Readmissions

Data Analytics

Dashboard shows readmissions spike for patients 65+ with diabetes discharged on Fridays.

→

Data Science

removing outliers/redundancy, fill out empty data
Cleans EHR data, tests causal factors, and builds a model that scores each patient's readmission risk.
extracts information from data and builds a model

→

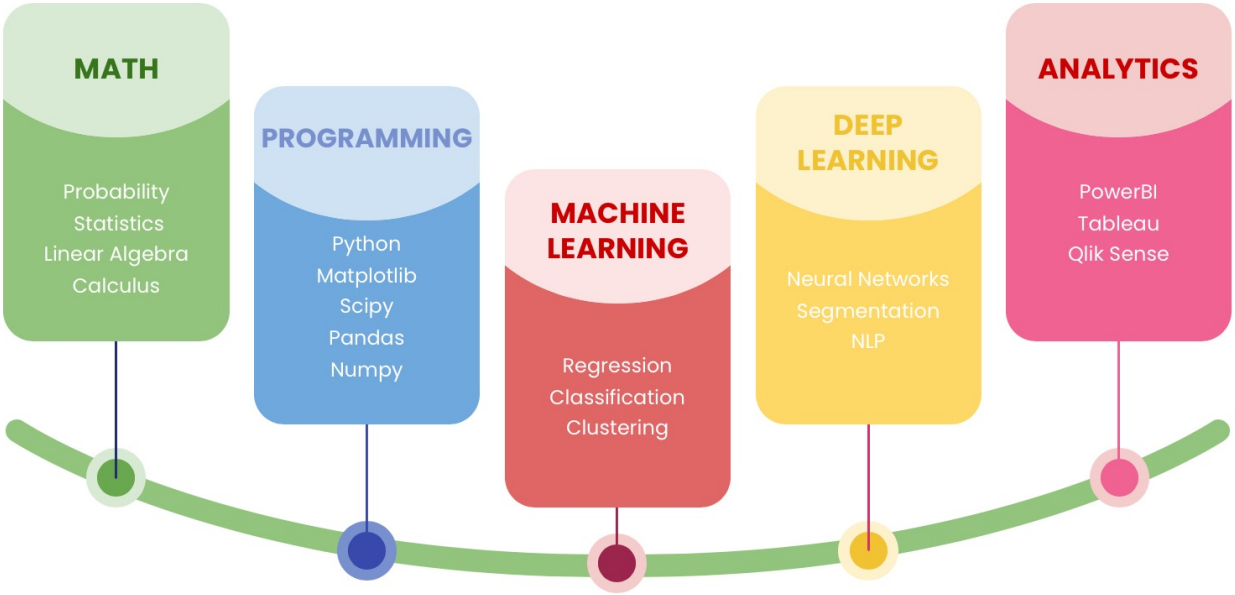
AI

Model is deployed live: flags high-risk patients in real time and auto-triggers a follow-up care plan.

"Data science builds the model; AI is the model doing something in the world."

DATA SCIENCE TOOLS

Tools you will need in data science



Data Science Process

DATA SCIENCE WORKFLOW

WHAT IS DATA SCIENCE?

01
BUSINESS UNDERSTANDING

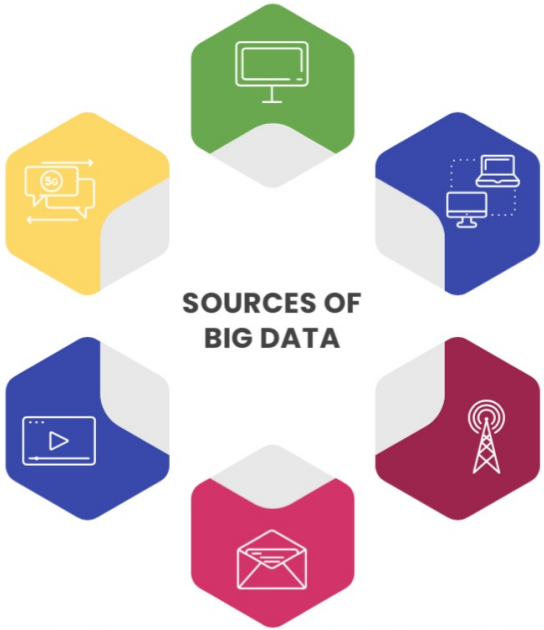
Identify the problem that must be considered in the study

02
DATA COLLECTION

Collect data that serves your study's objectives

03
DATA CLEANING

Fix the consistency in the data and handle missing values



04
FEATURE ENGINEERING

Transform your raw data into relevant and meaningful features

05
PREDICTIVE MODELS

Build Models
Train machine/deep learning models, and evaluate their performance and use them to make predictions

06
DATA VISUALIZATION

Communicate the finding with stakeholders and illustrate them with interactive visualization

needed for activities

DATA SCIENCE WORKFLOW

HOSPITAL READMISSION EXAMPLE

01

BUSINESS UNDERSTANDING

setting a goal

Frame the goal: predict which patients will be readmitted within 30 days.

02

DATA COLLECTION

health records from patients thru vital signs, age

Pull EHR records, discharge notes, and vitals for past patients.

03

DATA CLEANING (pre processing, before processing with data)

Fix missing lab values, standardize units, remove duplicate records.

we can have duplicates since we collect from multiple sources

we can have missing data since the hospital forgot to add records

04

FEATURE ENGINEERING

Some feature dont contribute to the output so its useless adding it to the analysis. or combining 2 features into one. or deducing a new factor from known information

Build risk flags from age, diagnoses, and discharge-day patterns.

05

PREDICTIVE MODELS

building the model by training it to get the model (mapping between input and output) to see if the patient is at risk or not

Train and evaluate a model that scores each patient's readmission risk.

06

DATA VISUALIZATION

can be done before prediction to know if the model is linear or any type of model

Build a dashboard flagging high-risk patients for the care team.

After building the model we use AI to improve outcomes

Data Science Process

1. **Data Collection:** Gathering raw data from various sources (databases, APIs, sensors, etc.).
2. **Data Cleaning:** Handling missing data, outliers, and formatting issues.
3. **Data Exploration:** Analyzing data patterns, distributions, and anomalies through descriptive statistics and visualizations.
4. **Feature Engineering:** Transforming raw data into meaningful inputs for machine learning models.
5. **Model Building:** Developing predictive or inferential models using statistical learning methods.
6. **Model Evaluation:** Measuring model performance through metrics like accuracy, precision, recall, etc.
7. **Deployment & Monitoring:** Integrating the model into production and continuously monitoring performance.

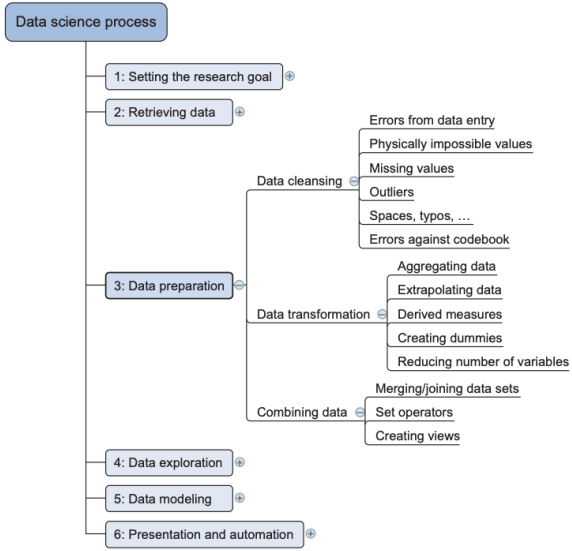


Figure 2.4 Step 3: Data preparation

Reference: https://livebook.manning.com/book/introducing-data-science/chapter-2/59

Course Data Analytics

Missing Data

Quiz1	Assignment:1	Quiz: 2	Assignment: 2	Assignment: 3	Programming Assignment total (Real)	Assignment: 4	Assignment: 5	Project total (Real)	Assignment: 6	Major Exam total (Real)	Course total (Real)	Final Exam: OpenCV	Final Exam TF	Final Exam ROS	Final Total	Total
7.65	10	9.11	10	10	18.71	18.5	10	19.25	19.5	19.5	58	15	14	9	38	96
8.77	10	8.67		10	18.97	20	10	20	19.75	20	59	11	10	5	26	85
8.89	10	9.33	10	10	19.29	18.5	10	19.25	17	17	56	12	12	5	29	85
2.41	10	-	10	10	16.2	20	10	20	11	11	48	10	8	4	22	70
7.16	10	10	10	10	18.86		10	20	17	17	56	14	13	7	34	90
7.53	10	7.56	10	10	18.03	20	10	20	16	16	55	11	11	6	28	83

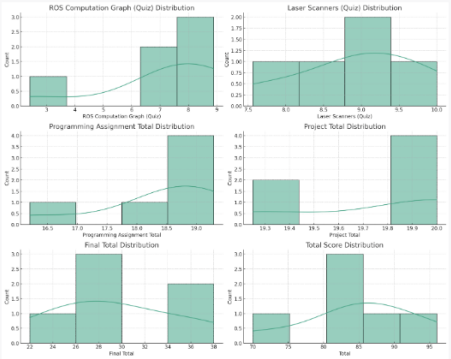
Exploratory Data Analysis (EDA)

From the initial statistics:

- The "Laser Scanners (Quiz)" column has a missing value since its count is 5 instead of 6.
- Most assignments have a consistent score of 10 for every student, indicating either they were easy or everyone did well.
- The total scores vary between 70 and 96, indicating a wide range of performances among the students.

Next, let's visualize the data distribution for significant columns using histograms and box plots.

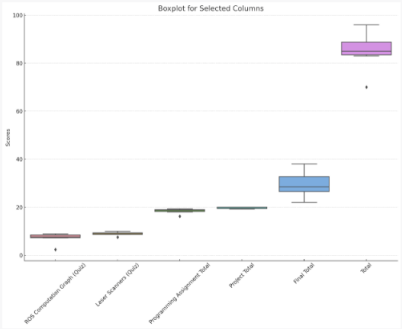
Finished working Show work



Here are the histograms showcasing the distribution of scores for selected significant columns:

Next, we'll use box plots to visualize the spread of scores and potentially identify any outliers.

Finished working Show work



Here are the box plots for the selected significant columns:

1. **ROS Computation Graph (Quiz)**: There's an outlier on the lower end, likely corresponding to the low score we observed in the histogram.
2. **Laser Scanners (Quiz)**: The scores are mostly between 8 and 10, but there's a lower whisker that might indicate an outlier or just a lower score.
3. **Programming Assignment Total**: The interquartile range (IQR) is narrow, indicating that most students scored similarly, close to the maximum.
4. **Project Total**: The scores are very concentrated at the upper limit. This reaffirms our observation that most students achieved the maximum score.
5. **Final Total**: The distribution is wider, with scores ranging from the low 20s to the upper 30s.
6. **Total**: There's a spread from 70 to 96, which means students' overall performance varied considerably.

There was an error generating a response

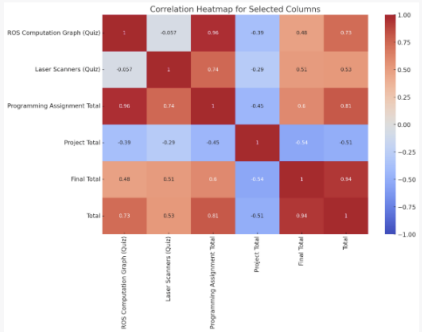
Regenerate

From the above results:

- Most students performed above average in the ROS Computation Graph Quiz and the Programming Assignments.
- Only half of the students scored above the average in the Laser Scanners Quiz.
- Only a third of the students performed above average in the final total.

Lastly, let's proceed with the Correlation Analysis to determine if there's a strong correlation between any two significant columns.

Finished working Show work



Exploratory Data Analysis (EDA)

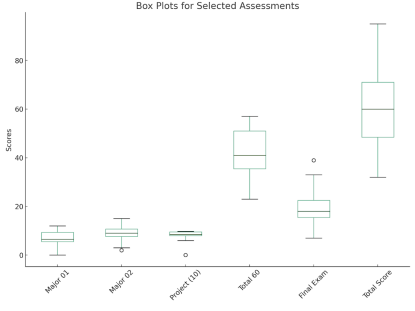
- **Purpose**
  - To understand data characteristics before modeling.
- **Techniques**
  - **Descriptive Statistics:** Mean, median, variance, etc.
  - **Data Visualization:** Histograms, scatter plots, box plots, etc.
  - **Correlation Analysis:** Identifying relationships between variables.
- **Tools**
  - Pandas, Matplotlib, Seaborn (Python); ggplot2, dplyr (R).

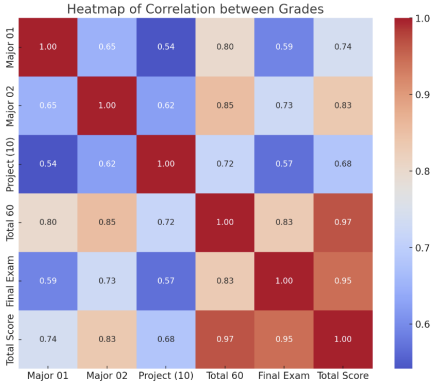
Analysis of Student Performance in CS102

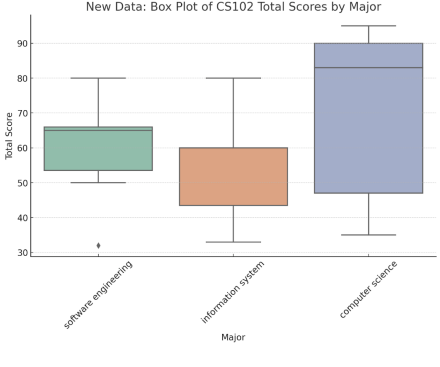
Summary Report:

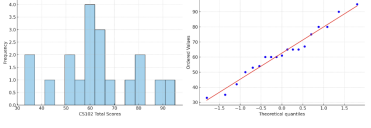
Analysis of Student Performance in CS102 - Section 780 Fall 2023

the considerable range and variance across assessments highlight the need for targeted support and interventions for students at different performance levels.

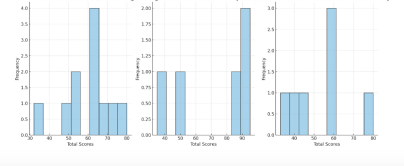


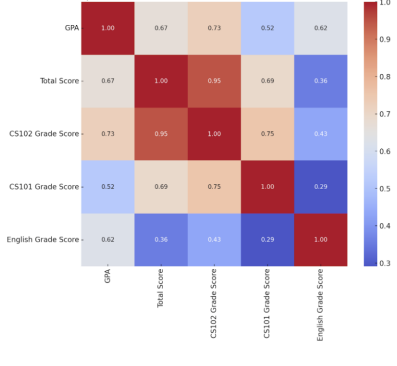






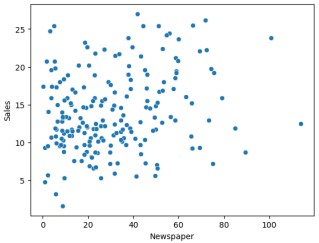
6. Major-Specific Observations:

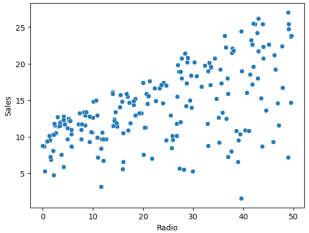


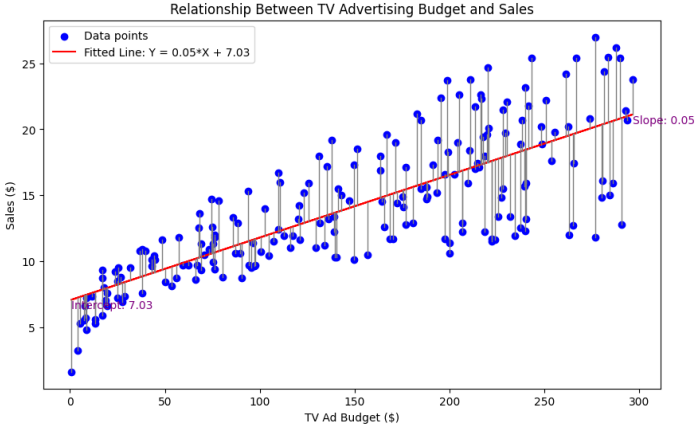


Advertising vs Sales - Predictive Modeling

	ID	TV	Radio	Newspaper	Sales
0	1	230.1	37.8	69.2	22.1
1	2	44.5	39.3	45.1	10.4
2	3	17.2	45.9	69.3	9.3
3	4	151.5	41.3	58.5	18.5
4	5	180.8	10.8	58.4	12.9

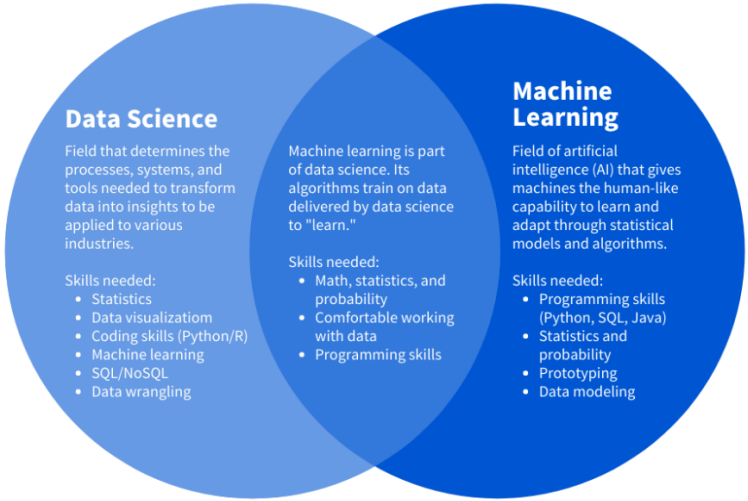






Intersection AI-Data Science

- **Huge overlaps in application of**
  - Machine Learning
  - Data driven Modelling
- **Data Science contributions to AI**
  - Data collection, management, evaluation, application
  - Data literacy
- **AI contributions to Data Science**
  - Increased perceptual modelling
  - Complex cognitive task solving
  - Better predictions
  - High-level information extraction

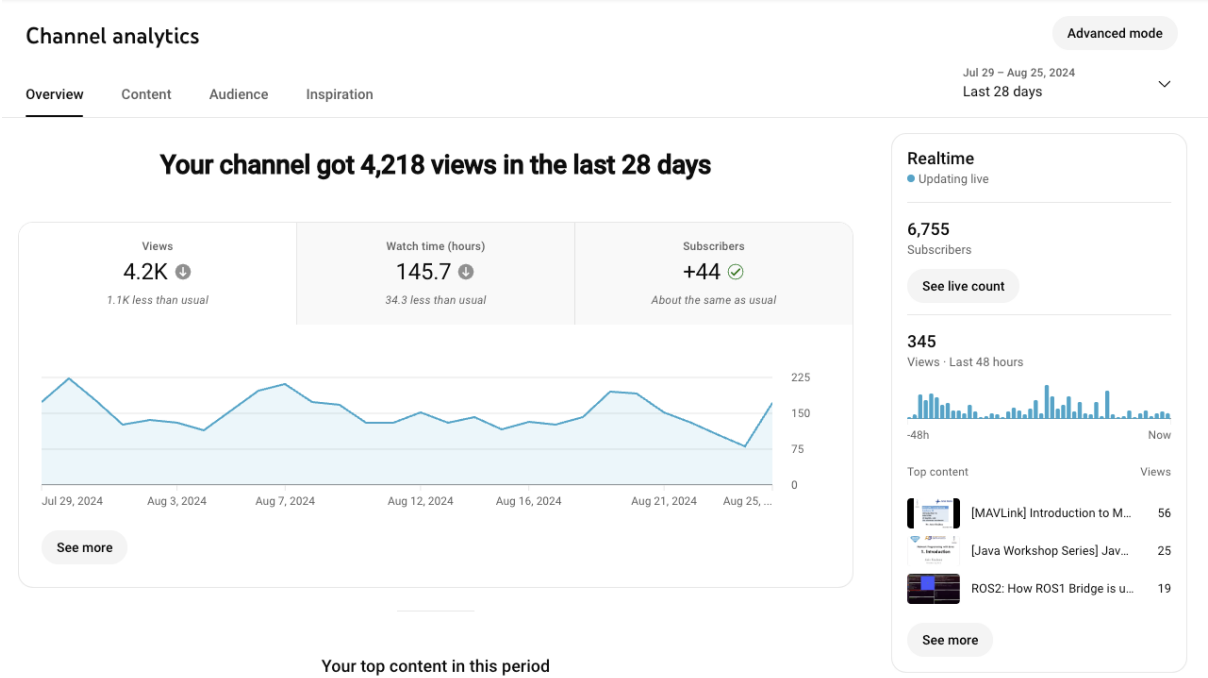


https://www.coursera.org/articles/data-science-vs-machine-learning?utm_medium=sem&utm_source=gg&utm_campaign=B2C_EMEA__coursera_FTCOF_career-academy_pmax-multiple-audiences-country-multi-set2&campaignid=20882109092&adgroupid=&device=c&keyword=&matchtype=&network=x&devicemodel=&adposition=&creativeid=&hide_mobile_promo&gad_source=1&gclid=Cj0KCQiAvbm7BhC5ARIsAFjwNHvWhfbn6PIQLceTX7272Df242FRF5crSSpodEVjoN3kfTmzWOSjsaIaAoKgEALw_wcB

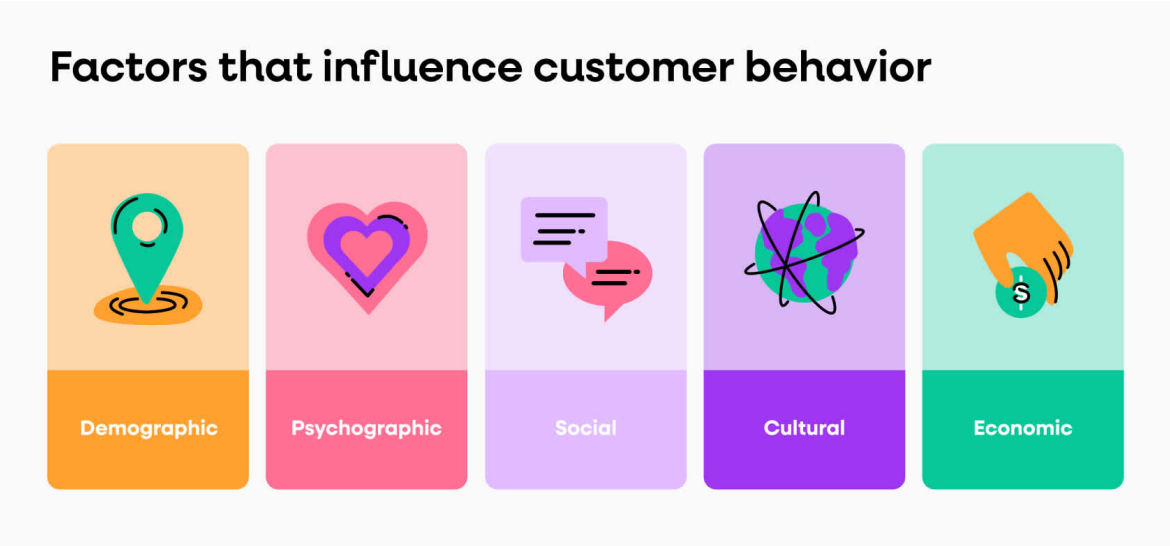
DATA ANALYTICS

TYPES

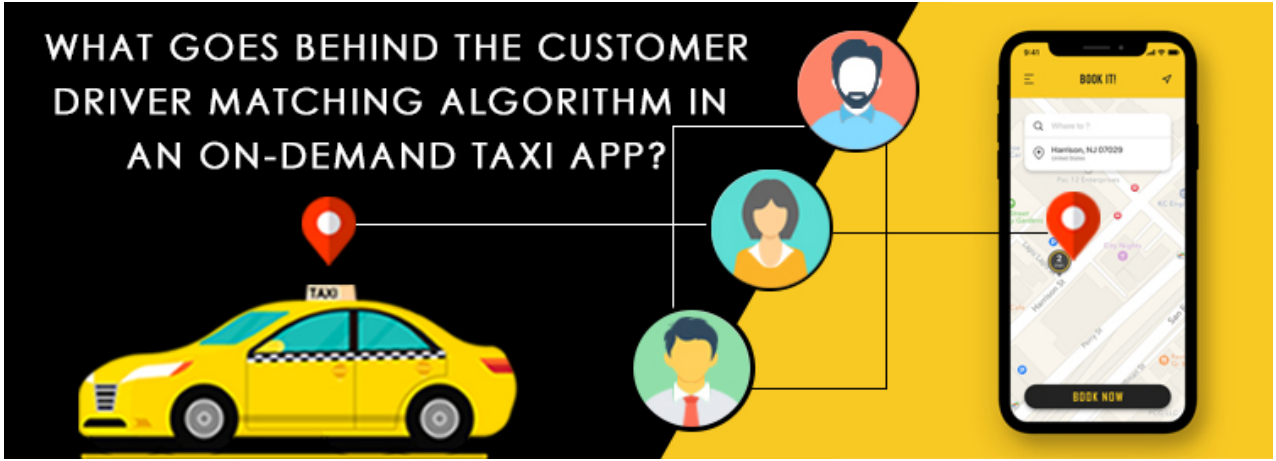
YouTube - Descriptive Analytics



Purchase Forecast - Predictive Analytics



Driver-Customer Matching - Prescriptive Analytics



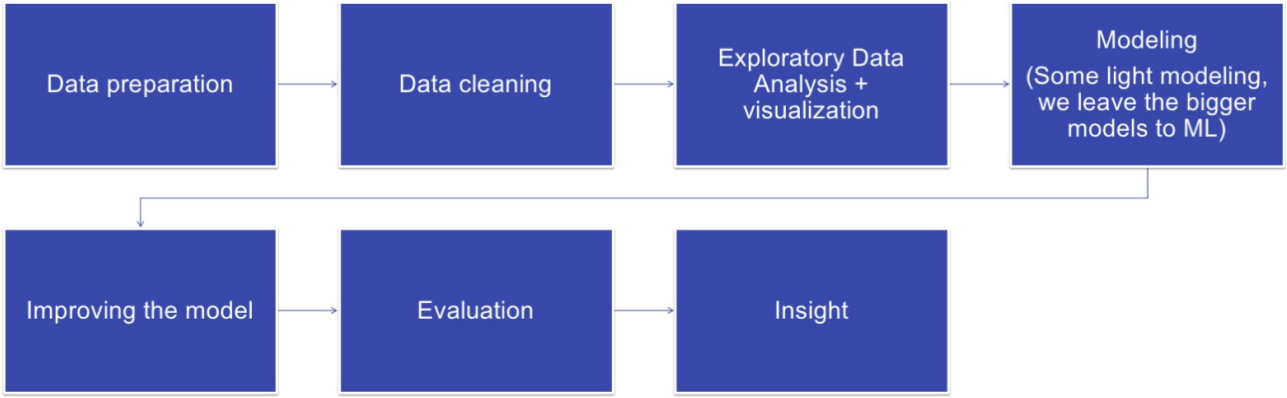
How to teach Data Science?

You need multiple courses:

- Programming (Java 1 and Java 2 set the stage)
- Linear Algebra
- Statistics and probability theory
- Databases & SQL
- Machine learning
- Big Data Analytics

What is this course?

We follow a similar process:



SUMMARY

We will start by looking at exploratory data analysis, using Python, next lecture in sha’ Allah.